Alexander Jacob

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Education

University of Southern California Los Angeles, CA
Bachelor of Science in Electrical and Computer Engineering *Expected May 2030*

Experience

Programming Sub-Team Member Sep. 2022 - June 2026
FIRST Robotics - Shaker High School *Latham, NY*
- Worked with the programming sub-team to test robot motors and sensors, verify component response, and troubleshoot hardware behavior.
- Member of a FIRST Robotics team that advanced to the FIRST Championship.

Tech Tutor & Volunteer June 2022 - Sep. 2025
William K. Sanford Library *Albany, NY*
- Taught seniors and community members to use computers, phones, tablets, email, texting, and internet tools at the library and local nursing homes.
- Troubleshot Wi-Fi and device issues and advised patrons on computers suited to their performance needs and budgets.
- Supported reading programs, children's activities, community events, and festival booths through setup and visitor service.

Front of House Team Member Apr. 2026 - July 2026
Chick-fil-A *Latham, NY*
- Served thousands of guests in a fast-paced environment by taking and bagging orders, preparing drinks and desserts, delivering food, and maintaining dining areas.
- Coordinated with coworkers to maintain speed and accuracy during busy periods while helping guests understand menu options and feel welcome.

Projects

Mechanical Foosball | *VEXcode, VEX V5, Distance Sensors, 3D Printing* 2025 - 2026
- Built a fully functional two-player game with 8 buttons, 6 motors, 2 distance sensors, 3D-printed players, and automatic first-to-3 scoring.
- Programmed motor controls, direction reversal, 125 mm goal detection, duplicate-count prevention, score tracking, and winner display in VEXcode.

Automated Connect 3 | *C++, Arduino Mega, Servos, CAD, 3D Printing* 2024 - 2025
- Designed an accessible 6x4 game using an Arduino Mega, 8 micro servos, 6 column-selection buttons, and 24 3D-printed pieces.
- Wrote C++ using a 2D array to validate moves, alternate players, actuate servos, detect wins and draws, and print game state.
- Modeled and printed about 1,230 g of custom parts; remeasured the board and redesigned ball geometry after testing exposed a fit problem.

Additional Autonomous Systems | *VEXcode, Motors, Sensors* 2022 - 2026
- Developed a cargo-handling robot with a 360-degree turntable and two-motor lifting arm and a four-stage pizza line using a conveyor, distance sensors, and a limit switch.

Leadership & Involvement

President, AI Club Sep. 2025 - June 2026
Shaker High School *Latham, NY*
- Led 20+ meetings on AI developments, applications, and ethics for members with varied technical backgrounds.
- Managed officers and project leads, funded collaborative activities, coordinated with other organizations, and grew the club to 30 active members.

Student Advisory Board Member 2023 - 2026
Shaker High School *Latham, NY*
- Represented student perspectives in discussions with school leaders and contributed to a pilot support system for middle school students.

Technical Skills

Languages: Python, Java, C/C++
Developer Tools: Git/GitHub, VS Code, Arduino IDE, VEXcode, Autodesk Inventor, AutoCAD
Hardware: Arduino Mega, VEX V5, servos, distance sensors, motors
Technical: CAD modeling, 3D printing, OpenCV basics, robotics programming, testing, troubleshooting, data collection